Tingting Du

UW-Madison | tdu35@wisc.edu | GitHub | LinkedIn | Google Scholar | Website

Interest: Robot Learning; Human-Robot Collaboration.

EDUCATION

University of Wisconsin-Madison — **Madison, Wisconsin, USA**
B.S. in Computer Science; — **Jan. 2025 ∼ May. 2027**
- **GPA:** 3.86/4.00; Dean's List; Autonomous Robotics, Operating Systems, Neural Networks.

University of California, Berkeley — **Berkeley, California, USA**
Visiting Student in Computer Science; — **Aug. 2023 ∼ Jun. 2024**
- **GPA:** 3.77/4.00; Global Access Scholarship; Real Analysis, Data Structures, Algorithms, AI.

Ningbo University — **Ningbo, Zhejiang, China**
Undergraduate in Linguistics; — **Sep. 2021 ∼ Jun. 2023**
- **GPA:** 3.80/4.00; Zhejiang Provincial Scholarship (Top 3%, 2×).

RESEARCH

Trimanual Manipulation — **RT^2 Lab, University of Wisconsin-Madison**
Advisor: Prof. Mike Hagenow — **Aug. 2026 ∼ Present**
- Learning from demonstration for trimanual human-robot team manipulation;
- Wired a unified emergency-stop interlock tying all three arms to a single safety circuit.

Vision-Language-Action and World Action Models — **CASE Lab, University of Maryland**
Advisor: Prof. Ang Li — **Nov. 2025 ∼ Aug. 2026**
- Multi-layer shared projector bridging 2D/3D representations in VLA models (NeurIPS 2026);
- Geometry-enhanced world action model (Under Review);
- Deployed and evaluated both on real-world manipulation with a UFACTORY xArm.

Student Modeling for Question Generation — **DM^2 Lab, University of Notre Dame**
Advisor: Prof. Meng Jiang — **Jul. 2024 ∼ Jan. 2025**
- Developed a student modeling approach for question generation (ACL 2025).

Situated Language Use and Convention Formation — **BAIR, University of California, Berkeley**
Advisor: Prof. Alane Suhr — **Feb. 2024 ∼ Jun. 2024**
- Analyzed ad-hoc convention formation in a large 3D-environment dialogue corpus.

PUBLICATIONS

- **T. Du**, Z. Wang, G. Sun, A. Li, **"XGenAct: Geometry-Enhanced World Action Models through Cross-Task Generation,"** arXiv:2610.03516, 2026.
- G. Sun, **T. Du**, K. Feng, C. Luo, Z. Shen, et al., **"ROCKET: Residual-Oriented Multi-Layer Alignment for Spatially-Aware VLA Models,"** *NeurIPS*, 2026.
- Z. Wang, B. Wang, H. Zhang, **T. Du**, T. Chen, et al., **"Vision-Language-Action in Robotics: A Survey of Datasets, Benchmarks, and Data Engines,"** *TMLR*, 2026.
- X. Tang, T. Qin, T. Peng, Z. Zhou, D. Shao, **T. Du**, et al., **"Agent KB: A Hierarchical Memory Framework for Cross-Domain Agentic Problem Solving,"** *ICML Workshop on Collaborative and Federated Agentic Workflows*, 2025.
- B. Nguyen, **T. Du**, M. Yu, L. Angrave, M. Jiang, **"QG-SMS: Enhancing Test Item Analysis via Student Modeling and Simulation,"** *ACL*, 2025.
- N. Tomlin, N. Zhou, E. Fleisig, L. Chen, T. Wright, L. Vinh, L.X. Ma, S. Eisape, **T. Du**, et al., **"Characterizing Language Use in a Collaborative Situated Game,"** arXiv:2512.03381, 2025.

SKILLS

- **Robotics:** ROS 2, UFACTORY xArm, multi-arm workcell bring-up and safety integration, Onshape CAD, 3D printing.
- **Machine Learning:** PyTorch, HuggingFace Transformers, vLLM, PEFT/LoRA, RLBench.
- **Languages & Infra:** Python, C/C++, Java, Julia, R, JS/TS, SQL, Assembly, LaTeX; Linux, Docker, AWS.